

Comprehensive Kinematic Analysis for Optimal Performance of a 6-DOF Robotic Manipulator with Prismatic Joint (RRRRRP)

M. Y. Alwardat^{a, *}, H. M. Alwan^b, and O. V. Kochneva^a

^a Peter the Great St. Petersburg Polytechnic University, St. Petersburg, Russia

^b University of Technology, Baghdad, Iraq

*e-mail: moh@alwardat@yahoo.com

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Abstract—The research is centered on the kinematics of six-degree-of-freedom robotic manipulators, which play a crucial role in the movement of various robotic joints. The kinematics issue is a conversion between Cartesian and joint space and vice versa. The study aims to develop a mathematical model for a robotic manipulator's forward and inverse kinematics with 6-DOF, including a prismatic joint. By utilizing a Denavit–Hartenberg representation, the forward and inverse matrices were calculated. The analytical solution for the kinematics of a 6-DOF manipulator enables an exploration of the manipulator's movement between two different configurations. The robotic manipulator includes six components: five revolute joints and one prismatic joint.

Keywords: robot manipulator, direct kinematics, inverse kinematics, Denavit–Hartenberg method

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INTRODUCTION

The field of robot kinematics involves the detailed investigation of the motion of robots (kinematics). During a kinematic analysis, all links' position, velocity, and acceleration are determined without considering the forces responsible for this motion. The main point of interest in robotics research involves the development of accurate inverse kinematics models.

Two kinematic issues are examined in this paper. The forward kinematics issue involves determining the coordinates and orientation of the gripper for a specific link configuration. Conversely, the inverse kinematics problem involves determining links that align the gripper with a given position and orientation. Several methodologies have been developed to solve the inverse kinematics problem. Among them are closed-form solutions, which encompass algebraic and geometric approaches. Algebraic methods are preferred over numerical ones due to faster computational speeds and the fact that they can guarantee the existence of a solution. An algebraic solution was provided for a robotic manipulator's inverse kinematics (IK) with six degrees of freedom and a prismatic joint. There are revolute joints (represented by a “R”) and prismatic joints (represented by a “P”) in the robot's kinematic structure.

RELATED LITERATURE

The field of robot modeling and simulation has extensively investigated the subject of kinematics. Clothier et al. [1] proposed a geometric model to solve the challenge of determining the joint angles necessary for achieving autonomous positioning of a robotic system. Sahu et al. [2] derived a novel approach, known as quaternion algebra, for solving the forward kinematic problem.

Popovic et al. [3] developed a methodology for examining the arm's upper extremity motion. In another investigation, Wang et al. [4] provided a comprehensive analysis of the overall body kinematics of a radial symmetrical six-legged robot. In the paper, [5] presented a mathematical methodology for analyzing the kinematics of a humanoid robot. Cubero [6] presented an inverse kinematics (IK) model that aims to determine the values of all joint variables for a serial arm manipulator, regardless of its specific type. The foundation of this model was derived from the forward kinematic solution. Kuma [7] reported on developing and analyzing a virtual model robot, specifically a 6-DOF robotic manipulator. The study focused on establishing both forward and inverse kinematic solutions for the robot.

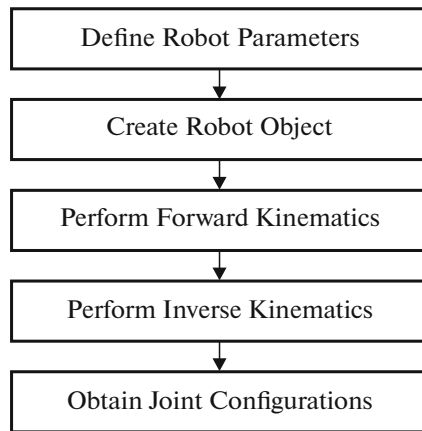


Fig. 1. Flow chart for design of robotic manipulator.

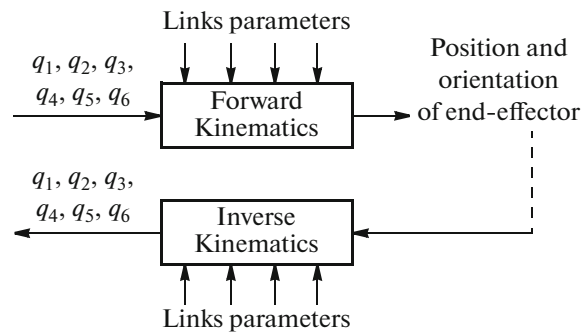


Fig. 2. Forward and inverse kinematics model q_1, q_2, q_3, q_4, q_5 and q_6 —joint angles.

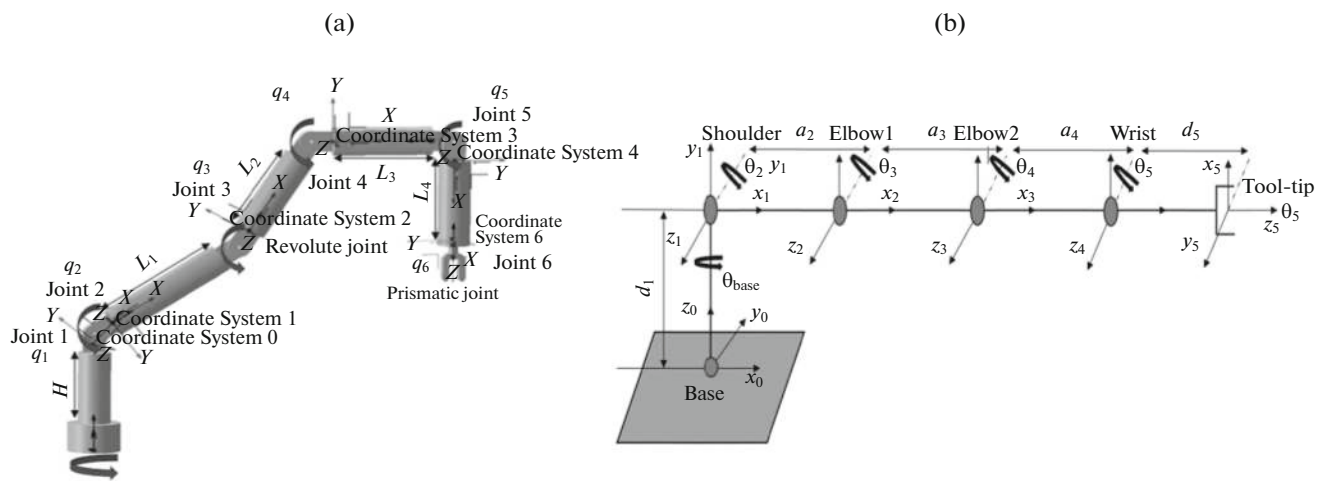


Fig. 3. Schematic of the robotic-manipulator (a), Kinematic scheme of the 6-DOF (b).

METHODOLOGY

System Modelling

The work scope has been segmented into several parts. The first step was to analyze the structural parameters of the 6-DOF robotic manipulator which contains a prismatic joint. then, we formed the D–H model and solved both the forward and inverse kinematic equations through algebraic techniques. The diagram illustrating the flow of the research methodology can be observed in Fig. 1.

Kinematic Analysis

The kinematic solutions for any robot typically consist of two main sub-problems: forward kinematics and inverse kinematics. Forward kinematics for a robot entails determining the position and orientation of its end-effector through the utilization of its joint [8–10]. On the other hand, inverse kinematics

involves computing joint angle values based on the end-effector's position and orientation [11–13]. The kinematic analysis model is shown in Fig. 2.

Structural Robotic Manipulator

The manipulator of six-DOF was built utilizing the SolidWorks software, as shown in Fig. 3a. This manipulator has five links (H, L_1, L_2, L_3, L_4), five angles (q_1, q_2, q_3, q_4, q_5) and one prismatic joint (q_6). The coordinate systems for the 6-DOF manipulator are set up using the DH method [14]. Figure 4b shows the required geometric parameters and notations.

DH Parameter

To conduct a kinematic analysis, the coordinate frames for each link were determined systematically, beginning at the base and ending at the end effector [15]. These frames were assigned numerical labels,

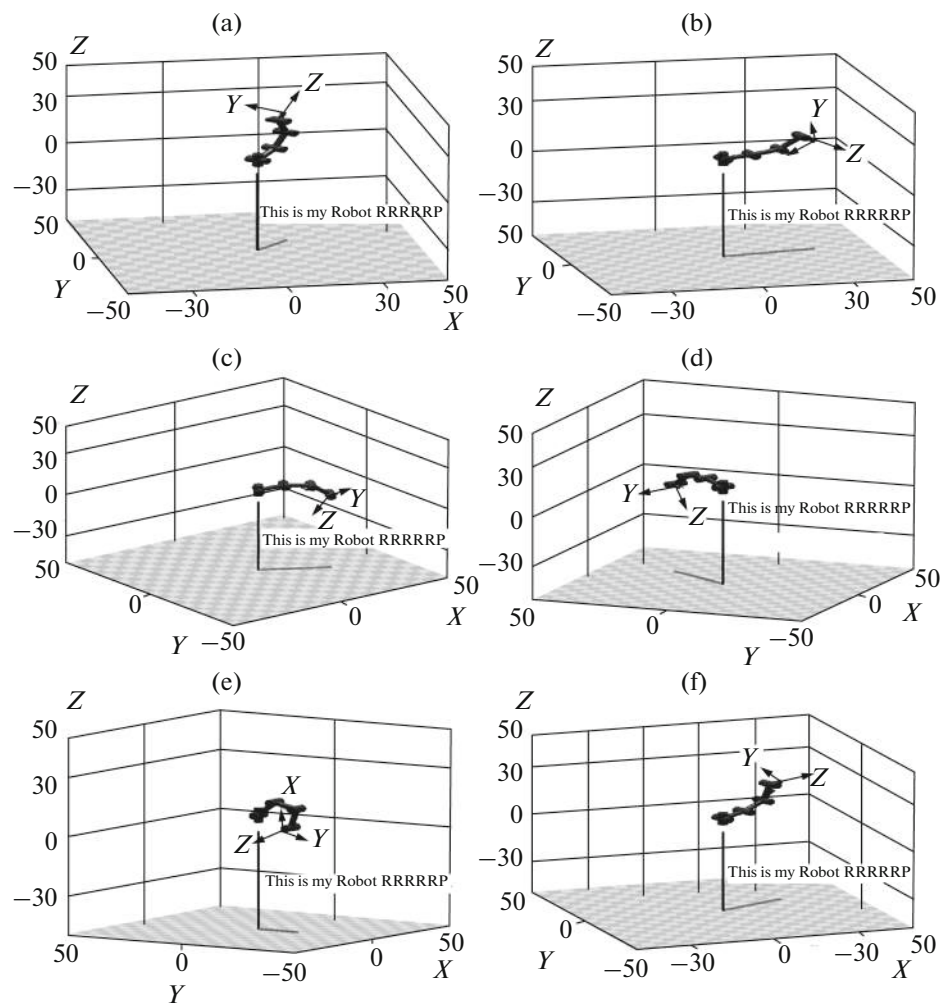


Fig. 4. Results of direct kinematics modeling: 1 (a), 2 (b), 3 (c), 4 (d), 5 (e) and 6 (f).

ranging from 1 to 6, to indicate their sequential order. In Table 1, the manipulator structural parameters were set out in detail. These parameters included link lengths, joint angles, and joint displacements.

The D–H parameters are corresponded to mounting configuration of links and determined based on the following rule [16]:

- q_i is the angle from x_{i-1} to x_i along z_{i-1} ;
- d_i is the distance from the intersection of z_{i-1} with x_i to the origin of $(i-1)$ system of axes;
- a_i is the shortcut between z_{i-1} to z_i ;
- α_i is the angle from z_{i-1} to z_i along x_i .

Forward Kinematic Analysis

The framework of the robotic manipulator is developed through the utilization of the D–H parameters and the MATLAB Robotics Toolbox. The fine (q) function is utilized for calculating the transformation matrix T and configuring the robotic manipulator based on the joint coordinates $q = [q_1 \ q_2 \ q_3 \ q_4 \ q_5 \ q_6]$. The form of the homogeneous transformation is given:

$$A_i = Rot_{z, \theta_i} Trans_{z, d_i} Trans_{x, a_i} Rot_{x, \alpha_i}.$$

In forward kinematics analysis, the joint angles are used to determine the end effector's position and ori-

Table 1. The D–H parameters of the manipulator

Joint i	α_{i-1} , deg	a_{i-1} , cm	q_i , deg	d_i , cm
0–1 1 Base	0	0	q_1	H
1–2 2 Shoulder	90	0	q_2	0
2–3 3 Elbow	0	L_1	q_3	0
3–4 4 Elbow	0	L_2	q_4	0
4–5 5 Wrist	0	L_3	q_5	0
5–6 6 Gripper	90	0	q_6	L_4

entation by substituting them into the homogenous transformation matrix between joints i and $i + 1$ [16].

$$A_i^{i+1} = \begin{bmatrix} c_{\theta i} & -s_{\theta i}c_{\alpha i} & s_{\theta i}a_{\alpha i} & a_i c_{\theta i} \\ s_{\theta i} & c_{\theta i}c_{\alpha i} & -c_{\theta i}s_{\alpha i} & a_i s_{\theta i} \\ 0 & s_{\alpha i} & c_{\alpha i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

by substituting the parameters in Table 1 to obtain on transformation matrices A_1^0 to A_6^5 , can be obtained as shown:

$$A_1^0 = A_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & H \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$A_2^1 = A_2 = \begin{bmatrix} c_2 & 0 & s_2 & 0 \\ s_2 & 0 & -c_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

where, the matrix A_1 shows the transformation between frame 1 to frame 0, $c_1 = \cos(q_1)$ and $s_1 = \sin(q_1)$

$$A_3^2 = A_3 = \begin{bmatrix} c_3 & -s_3 & 0 & c_3 L_1 \\ s_3 & c_3 & 0 & s_3 L_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$A_4^3 = A_4 = \begin{bmatrix} c_4 & -s_4 & 0 & c_4 L_2 \\ s_4 & c_4 & 0 & s_4 L_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$A_5^4 = A_5 = \begin{bmatrix} c_5 & -s_5 & 0 & c_5 L_3 \\ s_5 & c_5 & 0 & s_5 L_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$A_6^5 = A_6 = \begin{bmatrix} c_6 & 0 & -s_6 & 0 \\ s_6 & 0 & c_6 & 0 \\ 0 & 1 & 0 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Performing the composition from the i frame to the base frame we multiply from 1 to 6:

$$A_i^0 = A_1^0 A_2^1 A_3^2, \dots, A_i^{i-1} = \begin{bmatrix} R_n^0 & P_n^0 \\ 0 & 1 \end{bmatrix}.$$

Where: R is a 3×3 matrix for rotation and P is the position.

The total transformation A_6^0 between the base frame and the end-effector frame can be found by successive multiplication:

$$A_6^0 = A_1^0 A_2^1 A_3^2 A_4^3 A_5^4 A_6^5 = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (1)$$

Where $\{p_x, p_y, p_z\}$ represent the position and $\{(n_x, o_x, a_x), (n_y, o_y, a_y), (n_z, o_z, a_z)\}$ represent the orientation of the end-effector. The calculation of the orientation and position of the end-effector can be determined by using joint angles and the D-H parameters of the manipulator, as shown in Eq. (1).

The total transformation A_6^0 :

$$\Lambda_6^0 = \begin{bmatrix} c_{1234}c_{56} & s_{56} & c_{56}s_{1234} & s_{56}H + c_{56}[L_2s_{34} + L_1c_3] + L_3c_5 \\ s_{56}c_{1234} & -c_{56} & s_{1234}s_{56} & -s_{56}H + c_{56}[L_2c_{34} + L_1c_3] + L_3s_5 \\ s_{1234} & 0 & -c_{1234} & L_2s_{34} + L_1s_3 + L_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (2)$$

Inverse Kinematic Analysis

Solving inverse kinematics is more complex than the method of solving forward kinematics. Inverse kinematics is a technique for figuring out what values for the joints will result in the desired position and orientation of the end effector [17, 18]. By using the current coordinates and orientation of a robot, an inverse kinematics analysis is applied to calculate the joint angles, velocities, and accelerations required to manipulate the end effector and achieve the desired position [19]. This technique is fundamental when programming robots to carry out specific tasks. Using the algebraic solution technique, we have the ability to easily solve the matrix in Eq. (1) by:

$$A_6^0 = A_1^0 A_2^1 A_3^2 A_4^3 A_5^4 A_6^5. \quad (3)$$

To determine the value of q_i given specific numerical values for A_6^0 , it is necessary to perform multiplication on both sides by A_1^{-1}

$$A_1^{-1} \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = A_2^1 A_3^2 A_4^3 A_5^4 A_6^5.$$

We got:

$$c_1 p_x + s_1 p_y = L_1 c_3 c_{245} + L_2 c_4 + L_3 c_{45} + L_4 s_{245}, \quad (4)$$

$$-s_1 p_x + c_1 p_y = L_1 c_3 s_{245} + L_2 s_4 + L_3 s_{45} - L_4 c_{245}. \quad (5)$$

Table 2. Defined D–H parameters

Joint i	α_{i-1} , deg	a_{i-1} , cm	θ_i , deg	d_i , cm
0–1 1	0	0	q_1	$H = 8.5$
1–2 2	90	0	q_2	0
2–3 3	0	$L_1 = 10$	q_3	0
3–4 4	0	$L_2 = 10$	q_4	0
4–5 5	0	$L_3 = 10$	q_5	0
5–6 6	90	0	q_6	$L_4 = 10$

We can solve for c_5 as follows:

$$c_5 = \frac{p_x^2 + p_y^2 - (L_1 c_3)^2 - L_2^2 - L_3^2 - (L_4 s_{245})^2}{2L_2 L_3}.$$

We can also solve for s_5 as follows:

$$s_5 = \mp \sqrt{1 - c_5^2} \\ = \mp \sqrt{1 - \frac{(p_x^2 + p_y^2 - (L_1 c_3)^2 - L_2^2 - L_3^2 - L_4 s_{245})^2}{(2L_2 L_3)^2}}.$$

Square and add equations:

$$q_5 = A \tan 2(s_5, c_5).$$

Rearrange equations (7) and (8) to solve for c_1, s_1 :

$$c_1 = \frac{L_1 c_3 c_{245} + L_2 c_4 + L_3 c_{45} + L_4 s_{245} - s_1 p_y}{p_y}, \\ s_1 = \frac{L_1 c_3 s_{245} + L_2 s_4 + L_3 s_{45} - L_4 c_{245} - s_1 p_y}{p_x}, \\ q_1 = A \tan 2(s_1, c_1).$$

After solving for q_1 and q_5 , using the forward kinematics equations can make solving for q_3 and q_2 easy. From Eq. (2) we can find:

$$p_z - H = L_1 s_3; \quad s_3 = \frac{p_z - H}{L_1}; \\ q_3 = A \tan 2\left[\frac{p_z - H}{L_1}, \mp \sqrt{1 - \left(\frac{p_z - H}{L_1}\right)^2}\right].$$

Multiplying each side of Eq. (3) with $A_1^{-1} A_2^{-1}$:

$$A_1^{-1} A_2^{-1} \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = A_3^2 A_4^3 A_5^4 A_6^5. \quad (6)$$

Equating elements (3, 4) of the right-hand side matrix and the left-hand side matrix of Eq. (6), we can obtain:

$$s_2 p_x - c_2 p_y - H = L_4; \quad s_2 p_x - c_2 p_y = L_4 + H; \\ q_2 = A \tan 2(p_x, -p_y) \\ \mp \left[\sqrt{p_x^2 + p_y^2 - (L_4 + H)^2}, (L_4 + H) \right].$$

From Eq. (2) we can obtain

$$a_x = c_{56} s_{1234}, \quad a_y = s_{56} s_{1234}.$$

Dividing the two equations:

$$\frac{s_{56}}{c_{56}} = \frac{a_y}{a_x} q_{56} = A \tan 2(a_y, a_x),$$

and the we find:

$$q_6 = q_{56} - q_5.$$

Now multiply each side of Eq. (3) by:

$$A_1^{-1} A_2^{-1} A_3^{-1} \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = A_4^3 A_5^4 A_6^5. \quad (7)$$

Comparing the elements at positions 3 and 4 on both sides of Equation (7):

$$s_{23} p_x - c_{23} p_y - L_1 s_2 - H = L_4; \\ s_{23} p_x - c_{23} p_y = L_1 s_2 + H + L_4; \\ q_{23} A \tan 2(p_x, -p_y) \mp A \tan 2 \\ \times \left(\sqrt{p_x^2 + p_y^2 - (L_1 s_2 + H + L_4)^2}, L_1 s_2 + H + L_4 \right); \\ q_3 = q_{23} - q_2.$$

From the Eq. (2) we can also obtain:

$$a_z = -c_{1234}; \quad c_{1234} = -a_z q_{1234} \\ = A \tan 2\left(\sqrt{1 - a_z^2}, a_z\right); \quad q_4 = q_{1234} - q_3.$$

MODELING KINEMATICS

Modeling Forward Kinematic

The forward kinematics solution was created in MATLAB. For forward kinematic analysis, the joint angles were given as input. By using the D–H parameters shown in Table 2, we got the position of the end effector.

Table 3 contains the rotation angles $[q_1, q_2, q_3, q_4, q_5]$, position of the prismatic joints q_6 and the calculated positions of the end effector $[P_x, P_y, P_z]$. The results of direct modeling of kinematics are shown in Figs. 4a–4f.

The transformation matrices for cases 1 and 2 are equal:

$$T_1 = \begin{bmatrix} 0.0860 & -0.8985 & 0.4305 & 10.28 \\ 0.8554 & 0.2881 & 0.4305 & 10.28 \\ -0.5108 & 0.3312 & 0.7934 & 33.47 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \\ T_2 = \begin{bmatrix} -0.5565 & 0.1539 & 0.8164 & 31.81 \\ 0.4543 & 0.8792 & 0.1440 & 5.609 \\ -0.6956 & 0.4510 & -0.5592 & 14.49 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Table 3. Results of direct kinematics modeling

Case	Input (Joint angles)						Output (end effector position), cm		
	q_1 , deg	q_2 , deg	q_3 , deg	q_4 , deg	q_5 , deg	q_6 , cm	P_x	P_y	P_z
1	45	22.5	30	60	30	5	10.28	10.28	33.47
2	10	5	6	30	15	6	31.81	5.609	14.49
3	−30	15	−20	−40	20	2	22.38	−12.92	1.333
4	60	30	40	90	45	2	12.92	22.38	1.333
5	−90	45	−60	−90	45	4	0	−10.68	1.324
6	22.5	11.5	15	45	22.5	5	24.73	10.24	25.02

Table 4. Results of inverse kinematics modeling

Case	Input (End effector position), cm			Output (Joint angles)					
	P_x	P_y	P_z	q_1	q_2	q_3	q_4	q_5	q_6
1	10.28	10.28	33.47	45	36.7	25.59	17.8	62.28	0.3282
2	31.81	5.609	14.49	9.99	15.84	0	0.3	39.63	4.1978
3	22.38	−12.92	1.333	−30	18.86	−36.57	−28.45	21.17	0.3695
4	12.92	22.38	1.333	59.9	20.13	−40.6	−23.89	19.44	0.1064
5	0	−10.68	1.324	−90	−81.22	52.17	98.43	−129.37	3.5918
6	24.73	10.24	25.02	22.5	33.18	−0.022	−0.143	61.47	1.6461

The transformation matrices for cases 3 and 4 are equal:

$$T_3 = \begin{bmatrix} -0.3866 & 0.8465 & -0.3660 & 22.38 \\ 0.8514 & 0.4801 & 0.2113 & -12.92 \\ 0.3546 & -0.2299 & -0.9063 & 1.333 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$T_4 = \begin{bmatrix} -0.8514 & -0.4801 & -0.2113 & 12.92 \\ -0.3866 & 0.8465 & -0.3660 & 22.38 \\ 0.3546 & -0.2299 & -0.9063 & 1.333 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The transformation matrices for cases 5 and 6 are equal:

$$T_5 = \begin{bmatrix} 0.5440 & 0.8391 & 0 & 0 \\ 0.4195 & -0.2720 & 0.8660 & -10.68 \\ 0.7267 & -0.4711 & -0.5000 & 1.324 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$T_6 = \begin{bmatrix} -0.1474 & -0.3605 & 0.9210 & 24.73 \\ 0.5278 & 0.7589 & 0.3815 & 10.24 \\ -0.8365 & 0.5423 & 0.0785 & 25.02 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Modeling Inverse Kinematics

In inverse kinematic analysis, the end effector positions were given as input, and the joint angles were calculated. Table 4 shows the values of the end effector position $[P_x, P_y, P_z]$, as well as the corresponding rotation angles $[q_1, q_2, q_3, q_4, q_5, q_6]$, obtained using inverse kinematics for the 5RP manipulator. The end effector can be reached at the specified point; therefore, the joint angles $[q_1, q_2, q_3, q_4, q_5, q_6]$ are found successfully. The results of inverse modeling of kinematics are shown in Figs. 5a–5f.

RESULTS AND DISCUSSION

The paper analyzed a 6-DOF robotic manipulator with a prismatic joint and investigated its kinematic characteristics. Analytical methods were utilized in the investigation, employing the DH technique to derive forward and inverse kinematics solutions. The obtained joint angles are presented in Table 3, playing a vital role in calculating the inclusive homogeneous transformation matrix. This matrix encompasses position and orientation parameters, which are crucial for understanding the kinematic behavior of the robotic manipulator.

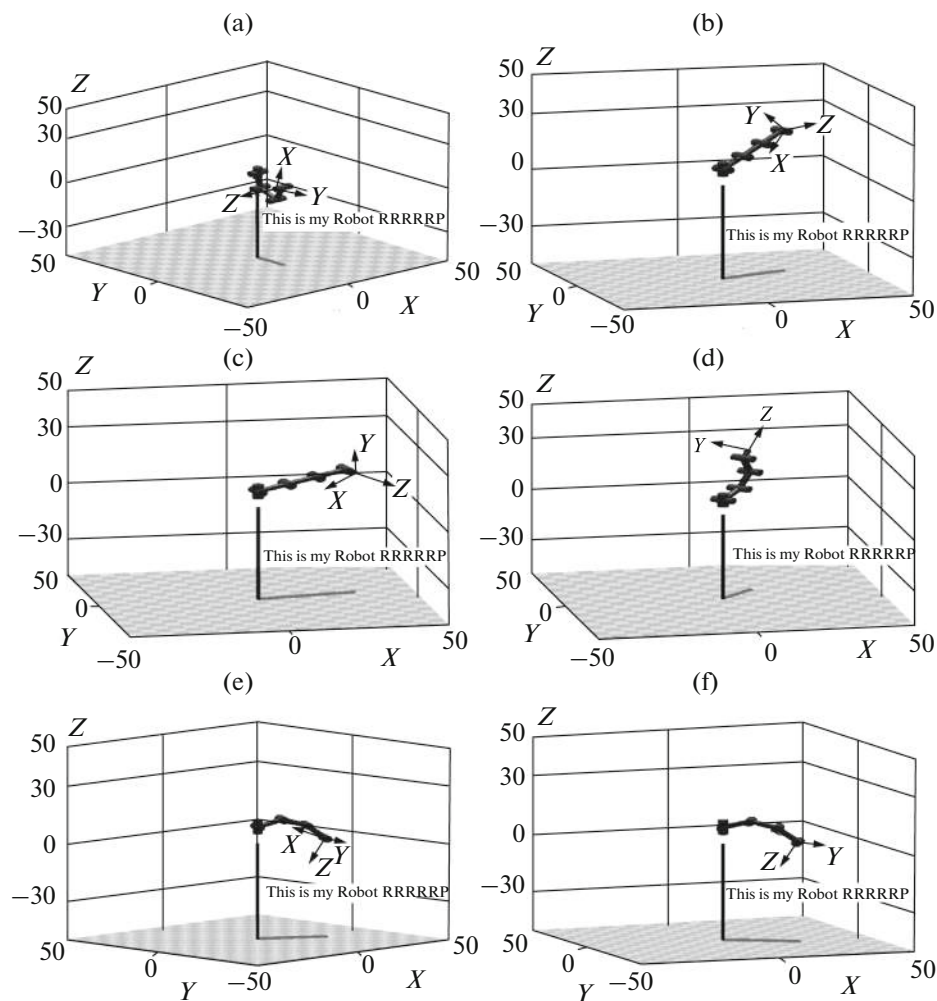


Fig. 5. Results of inverse kinematics modeling: 1 (a), 2 (b), 3 (c), 4 (d), 5 (e) and 6 (f).

CONCLUSIONS

(1) The current research provides an extensive mathematical solution to the forward and inverse kinematics of a robotic manipulator with six degrees of freedom, which includes a prismatic joint. The evaluation focuses on the kinematic analysis of a robotic manipulator with six degrees of freedom.

(2) The mathematical model of end effector position and orientation was solved in the MATLAB toolbox. The validation process confirmed the accuracy of the kinematic analysis and helped us understand the kinematics of a 6-DOF robotic manipulator.

(3) The generated mathematical solutions empower us to control the robot's movement efficiently, opening up opportunities for advanced applications across various industries.

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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